

Exploiting LES computations and machine learning to improve wind turbine wake reproduction in URANS

Andrea Carloni^a, Giovanni Delibra^a, Valerio Francesco Barnabei^a, Alessandro Corsini^a, Lorenzo Tieghi^b, and Filippo De Girolamo^c

^aSapienza University of Rome, Rome, Italy

^bTrento University, Trento, Italy

^cKTH Royal Institute of Technology, Stockholm, Sweden

Abstract

URANS computations are widely considered unreliable for reproducing wind turbine wake development, limiting their applicability in farm simulations where wake interactions are critical.

This work explores a pipeline combining k-means clustering, to automatically segment the wake into different regions at runtime as in [1, 2], and neural network regression to derive a machine-learned correction of eddy viscosity in URANS, based on high-fidelity LES data.

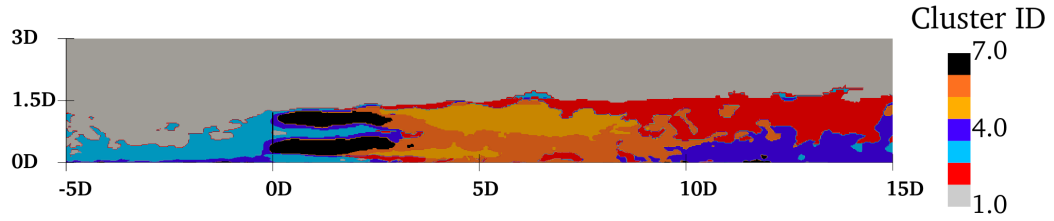


Figure 1: Wake decomposition via clustering [1].

LES, RANS, and ML-enhanced RANS simulations are performed in OpenFOAM-v22.12, modelling the NREL 5 MW reference turbine via the Actuator Line Method with a synthetic atmospheric boundary layer at the inflow.

A data-driven correction of RANS eddy viscosity is introduced as an additive term,

$$\nu_{t,ML} = \nu_{t,RANS} + \Delta\nu_{t,ML,i}, \quad i = 1, \dots, N_c, \quad (1)$$

Where N_c is the total number of clusters allowing region-specific eddy-viscosity corrections. Trained models are embedded in a modified PIMPLE solver, where the eddy viscosity is updated iteratively within the turbulence-correction loop.

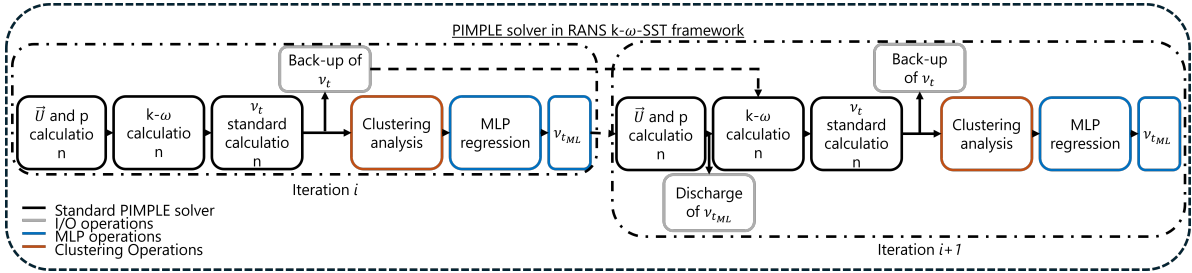


Figure 2: Modified PIMPLE solver pipeline.

Results show improved wake-flow prediction over baseline RANS in both near and far wake. The method retains accuracy on grids up to $7\times$ coarser than LES, yielding substantial computational savings. Non-dimensional topological features support interpretability and generalization to different flow conditions and turbine configurations.

References

- [1] Andrea Carloni, Giovanni Delibra, Filippo De Girolamo, Valerio Francesco Barnabei, and Lorenzo Tieghi. Data-driven decomposition of horizontal-axis wind-turbine wakes. In *ASME Turbo Expo 2026*, Milan, Italy, 2026.
- [2] Lorenzo Tieghi, Davide Cerbarano, Filippo De Girolamo, Valerio Barnabei, and Giovanni Delibra. A novel methodology for the automatic decomposition of hawt wakes with k-means clustering. *Wind Energy*, 28, 05 2025.